

NEWTON 11 SW, GA, USA

WMO: 723220

Lat: 31.1923N

Lon: 84.4465W

Elev: 48

StdP: 100.75

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 63829

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.1	-1.3	-10.5	1.5	3.1	-7.9	1.9	4.4	4.6	12.6	4.0	11.2	0.6	N/A	0.302

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	34.9	23.9	33.8	24.0	32.9	24.0	26.3	31.5	25.8	30.8	25.4	30.3	1.9	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.1	20.3	28.2	24.6	19.7	27.8	24.1	19.1	27.3	81.7	31.6	79.7	31.0	77.9	30.3	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	3.8	3.3	3.0	DB	-6.3	36.1	1.7	1.4	-7.5	37.1	-8.5	38.0	-9.5	38.8	-10.8	39.8
(5)				WB	-7.0	27.1	2.0	0.7	-8.4	27.6	-9.6	28.0	-10.8	28.4	-12.3	29.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.3	9.9	11.9	15.4	19.0	23.3	26.3	27.1	27.0	25.0	20.1	14.4	11.7
	DBStd	7.23	5.55	4.88	4.47	3.40	2.82	1.70	1.37	1.45	2.26	3.98	4.56	5.12
	HDD10.0	168	72	33	9	1	0	0	0	0	0	1	12	41
	HDD18.3	973	264	184	111	33	4	0	0	0	1	29	134	213
	CDD10.0	3565	70	87	177	269	411	490	532	528	451	312	144	93
	CDD18.3	1327	3	5	20	52	157	240	273	269	202	83	16	7
	CDH23.3	12007	11	35	174	506	1479	2275	2574	2422	1716	679	111	26
	CDH26.7	4911	0	2	22	109	577	1041	1180	1068	711	188	13	0
Wind	WSAvg	1.2	1.3	1.4	1.5	1.4	1.3	1.1	1.0	0.9	1.1	1.1	1.2	1.2
Precipitation	PrecAvg	1259	121	101	143	78	58	137	152	121	119	72	78	89
	PrecMax	1753	379	171	284	187	155	250	250	237	367	218	229	164
	PrecMin	830	24	15	9	16	3	38	92	56	21	9	11	22
	PrecStd	239	77	47	90	44	39	65	51	51	86	49	53	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.7	26.6	28.8	30.7	34.3	36.4	35.4	35.8	35.2	32.0	28.3	25.6
		MCWB	18.6	19.5	18.3	20.7	21.8	22.4	24.8	24.1	23.2	22.4	20.6	20.2
	2%	DB	22.8	24.3	27.1	29.4	32.6	34.9	34.2	34.1	33.2	30.2	26.3	23.9
		MCWB	17.8	17.8	18.3	20.0	21.4	23.3	24.6	24.7	23.1	20.7	19.2	18.9
	5%	DB	20.9	22.1	25.3	27.9	31.3	33.4	33.4	33.0	31.9	28.9	24.4	22.0
		MCWB	17.1	17.5	17.8	19.2	21.2	23.9	24.7	24.6	23.1	19.8	18.0	18.7
	10%	DB	18.9	20.3	23.6	26.5	30.1	32.0	32.3	31.9	30.8	27.3	22.6	20.3
		MCWB	15.9	16.5	16.7	18.3	20.9	23.6	24.7	24.7	23.0	19.5	17.5	18.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.4	20.8	20.9	22.8	24.2	26.3	26.8	27.1	25.8	25.0	23.1	22.0
		MCDB	22.5	23.9	24.6	27.3	29.1	31.8	31.9	32.5	30.6	28.6	25.6	24.0
	2%	WB	19.2	20.0	20.1	21.9	23.6	25.6	26.2	26.4	25.1	23.8	21.8	20.9
		MCDB	21.4	23.3	23.7	26.2	28.3	30.9	31.8	31.2	29.9	26.9	24.0	22.6
	5%	WB	18.0	18.9	19.2	21.1	23.0	25.1	25.7	25.9	24.7	22.7	20.4	19.9
		MCDB	20.3	22.0	23.1	25.4	27.6	30.3	31.1	30.5	29.4	26.0	22.8	21.8
	10%	WB	16.7	17.6	18.3	20.2	22.4	24.6	25.3	25.5	24.1	21.6	18.8	18.6
		MCDB	19.1	20.4	22.4	24.4	27.3	29.4	30.4	30.0	28.9	25.5	21.9	20.5
Mean Daily Temperature Range	5% DB	MDBR	11.9	12.1	12.7	12.8	12.2	10.7	10.3	9.9	10.7	12.4	13.2	11.6
		MCDBR	12.3	13.6	13.8	14.3	13.7	13.1	11.7	11.6	12.4	13.6	13.8	11.8
		MCWBR	7.5	8.0	6.4	5.9	4.5	3.9	3.7	3.8	4.4	6.2	7.4	7.5
	5% WB	MCDBR	10.2	11.2	11.3	11.0	10.7	10.4	10.5	10.1	10.4	10.3	10.1	9.7
		MCWBR	7.3	7.9	6.2	6.0	4.1	4.1	3.8	3.8	3.9	5.5	6.4	7.1
Clear-Sky Solar Irradiance	taub	0.327	0.334	0.355	0.385	0.434	0.482	0.508	0.496	0.426	0.374	0.351	0.333	
	taud	2.510	2.486	2.441	2.358	2.252	2.180	2.119	2.151	2.334	2.451	2.458	2.512	
	Ebn at Noon	902	928	928	907	860	815	791	797	845	871	864	875	
	Edh at Noon	88	98	110	124	139	149	158	151	121	101	92	84	
All-Sky Solar Radiation	RadAvg	3.04	3.61	4.80	5.89	6.46	6.06	5.99	5.50	4.97	4.32	3.41	2.65	
	RadStd	0.20	0.43	0.40	0.47	0.44	0.45	0.37	0.38	0.40	0.42	0.27	0.25	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.85	N/A	N/A	N/A	N/A	N/A	-48	-158	+259	+132
(47)	Regional (37 neighbors)	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+160	+102

Nomenclature: See separate page